

Exercise 13

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Introduction to Electrodynamics

ISBN: 9780321856562

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We need to find the electric field a distance s from an infinitely long straight wire. Consider a cylindrical Gaussian surface with radius of s and length of l as shown in the following figure. Apply Gauss's law,

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

the flux through the two sides of the cylinder is zero, since the flux lines are perpendicular to the "perpendicular line on the bases". The flux through the curved surface is constant, so on that surface we can pull the electric field out of the integral and integrate over that surface to get the total area of the curved surface, that is $2\pi sl$

$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot l = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

the enclosed charge equals the charge density λ multiplied by the length inside the surface l , that is $q = \lambda l$, so,

$$E \cdot 2\pi s \cdot l = \frac{1}{\epsilon_0} \lambda l$$

the direction of the field is radially outward from the wire, so,

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}$$

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